

Kyaw Soe Lwin

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PROFESSIONAL SUMMARY

Data Science undergraduate at UC San Diego (4.0 GPA) building leakage-free ML pipelines, Generative AI/LLM systems (RAG, PEFT/LoRA), and computer vision applications in Python. Proficient in Scikit-Learn, XGBoost, PyTorch, and SQL, with hands-on cloud deployment across AWS and GCP (Vertex AI).

EDUCATION

University of California, San Diego (UCSD) – San Diego, CA

Expected June 2028

Bachelor of Science in Data Science

Relevant Coursework: Advanced Machine Learning, Data Engineering, Statistical Modeling, Algorithmic Data Analysis, Linear Algebra, Multivariable Calculus, Data Structures & Algorithms, Object-Oriented Programming (Java)

TECHNICAL SKILLS

Statistics & Experimentation: A/B Testing, Hypothesis Testing, Statistical Data Analysis, Leakage Audits, Generalization Validation, Bayesian Optimization (Optuna), Time Series Analysis, Geospatial Analysis (Getis-Ord Gi*)

Data Science & Machine Learning: Scikit-Learn, XGBoost, PyTorch, TensorFlow/Keras, Supervised/Unsupervised Learning, SHAP Interpretability, Feature Engineering, ColumnTransformer Pipelines, CNNs, Computer Vision (OpenCV)

Programming & Data Systems: Python, SQL (Querying & Aggregations), Java, Bash/Linux, Pandas, NumPy, Exploratory Data Analysis, Data Auditing & Schema Standardization

GenAI, LLMs & Agents: Hugging Face Transformers, RAG Architectures, Chroma Vector DB, LLM APIs (Anthropic Claude), Prompt Engineering, PEFT/LoRA Fine-Tuning, RLHF

Cloud & MLOps: AWS, GCP (Vertex AI), Docker, Kubernetes (GKE), Streamlit, Git/GitHub, Agile/Scrum, Automated Unit Testing, CI/CD

AI-Assisted Development: Anthropic Claude Code, Google Antigravity, Agentic Coding Workflows

TECHNICAL PROJECTS

Telco Customer Churn Prediction & Profit Thresholding

2026

Python, SQL, Scikit-Learn, XGBoost, Optuna, SHAP, Streamlit · [GitHub Repo](#)

- Formulated a leakage-free predictive ML framework on 7,043 customer records, enforcing a stratified 70/15/15 train/val/test split to guarantee generalization.
- Executed a 30-trial Bayesian hyperparameter search (Optuna) with cross-validation, achieving 0.844 ROC-AUC / 0.671 PR-AUC with a <3.5-point generalization gap.
- Interpreted model decisions using TreeSHAP attribution and mapped findings to a cost-sensitive ROI matrix (\$20 cost / \$200 LTV) via an interactive Streamlit dashboard.

Priceout Collective — Affordability Policy Simulator

2026

Python, Pandas, XGBoost, Scikit-Learn, JavaScript · Building for Good Hackathon · [GitHub Repo](#)

- Modeled 1.17M household records across 4 policy dimensions, precomputing 945 scenario combinations for zero-latency, stakeholder-facing simulation.
- Diagnosed and eliminated target leakage in an XGBoost model, reaching 97.3% accuracy / 0.998 AUC, and validated results against ALICE and HUD CHAS housing benchmarks.

Local Expert — Offline PDF QA Engine (RAG Pipeline)

2026

Python, Hugging Face Transformers, Chroma, Anthropic Claude API, Streamlit · [GitHub Repo](#)

- Architected an end-to-end Retrieval-Augmented Generation system, hand-crafting chunking, sentence embeddings, and vector indexing to deliver citation-grounded answers over private documents.
- Designed a swappable LLM provider interface (Anthropic Claude API and local Ollama) behind a shared configuration layer.

LoRA TinyLlama-1.1B Instruction Fine-Tuning

2026

PyTorch, Hugging Face, PEFT/LoRA, Transformers, Google Colab · [GitHub Repo](#)

- Configured low-rank adaptation (LoRA) matrices targeting attention projection layers, reducing trainable parameter footprint by >95% on a resource-constrained T4 GPU.

EXPERIENCE & HONORS

Data Science Alliance – Data Science Volunteer

Sep 2026 – Present

- Built an end-to-end data auditing/validation pipeline on longitudinal unsheltered-homelessness data across Downtown San Diego (2012–present), cross-referencing multi-year counts to reach 97.5%+ fidelity.
- Standardized schemas and built geospatial crosswalks across 380+ downtown blocks, preparing datasets for time-series decomposition and spatial hotspot forecasting (Getis-Ord Gi*).

Teacher Ni Language Centre – Academic Tutor

Oct 2022 – Dec 2023

- Mentored 40+ students through structured technical curricula, delivering clear, actionable feedback via weekly evaluations.

Honors: Jack Kent Cooke Transfer Scholarship Semifinalist (2026) · Sterling Redman & F.L. Griffin Scholar (2025–2026)